

# Trained



Large labelled training set

train





Corpus-specialized features

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Standard



Cross-env.





## Standard



**Cross-env.**



# Training-free, prior work

Frozen patch features

Codebook or Segmentation CLS attention

Prior-selected / aggregated features

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Standard **×** Cross-env. —



## Standard



**Cross-env.**



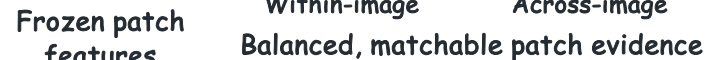
**Ours**

Frozen patch features → Within-image → Across-image

Within-image: Balanced, matchable patch evidence

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Standard ✓ Cross-env. ✓



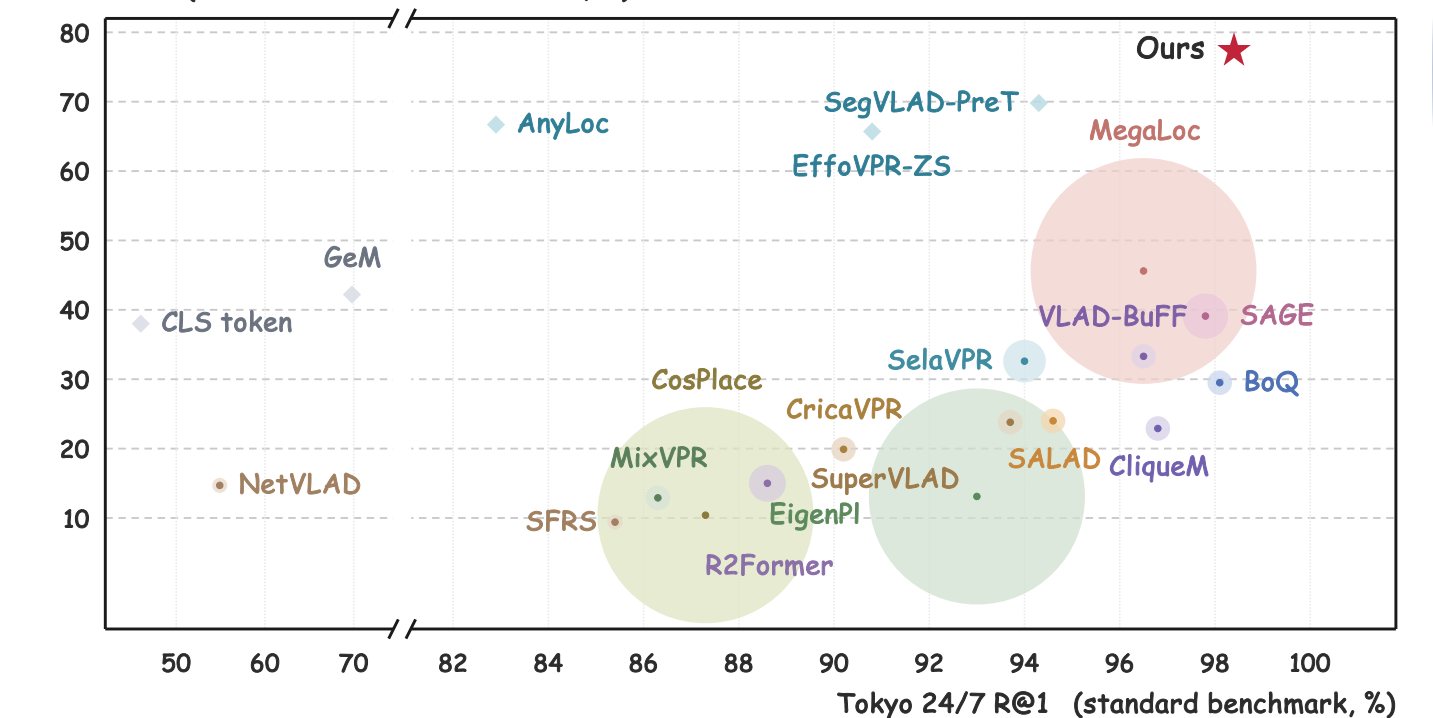
## Standard



**Cross-env.**



VP-Air R@1 (cross-environment benchmark, %)



★ ours (training-free)

- ▶ training-free

- **trained**

● 1 million images

5 million images

*standard*

Pitts250k  
100  
Pitts30k  
Tokyo 24/7  
MSLS-val  
Nordland  
St Lucia  
Oxford  
AmsterTime  
17 Places  
Baidu Mall  
Gardens Point  
VP-Air  
Nardo-Air  
Nardo-Air R  
Hawkins  
Laurel  
Mid-Atlantic

*cross-environment*

**Ours** **MegaLoc** **AnyLoc**

20  
60

*standard*

*cross-environment*

— Ours

— MegaLoc

## AnyLoc